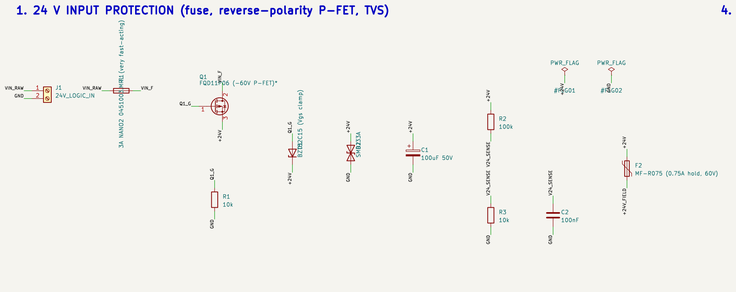
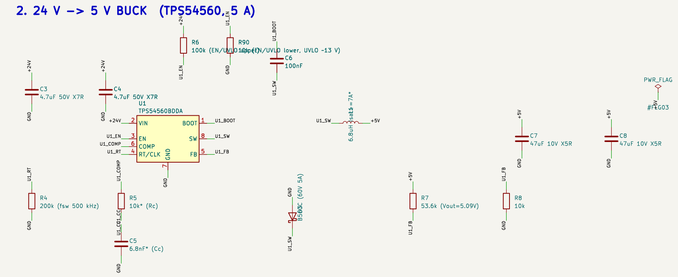
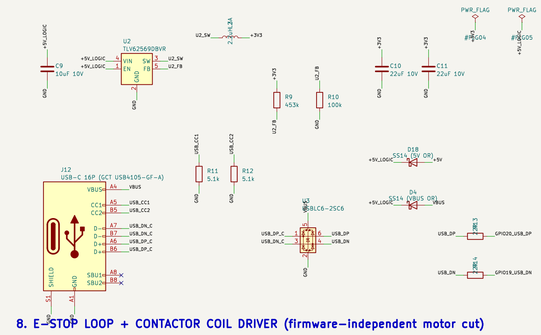
1. 24 V INPUT PROTECTION (fuse, reverse-polarity P-FET, TVS)



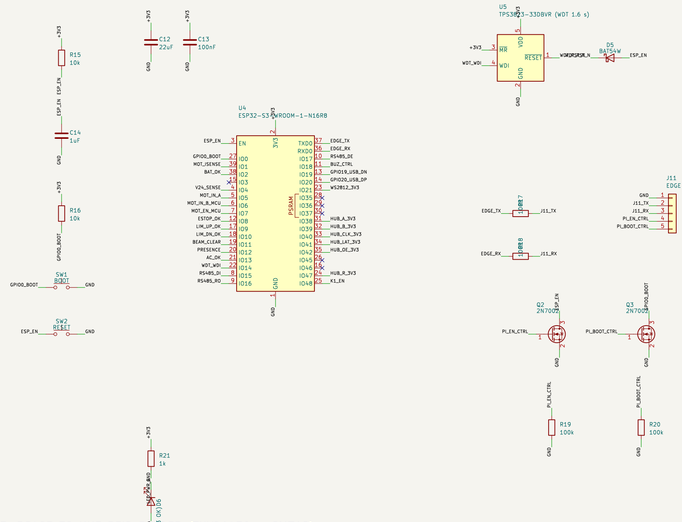
2. 24 V -> 5 V BUCK (TPS54560, 5 A)



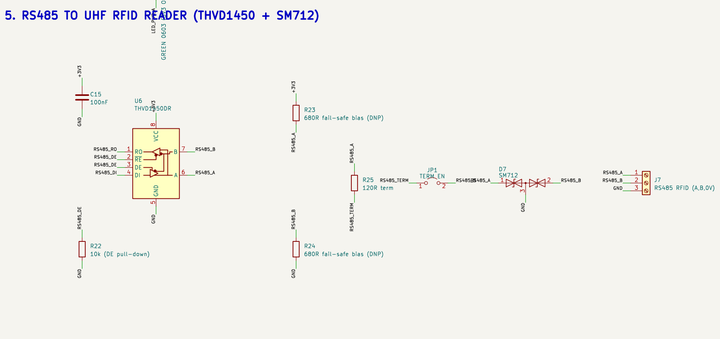
3. 5 V -> 3.3 V BUCK (TLV62569, 2 A) + USB-C



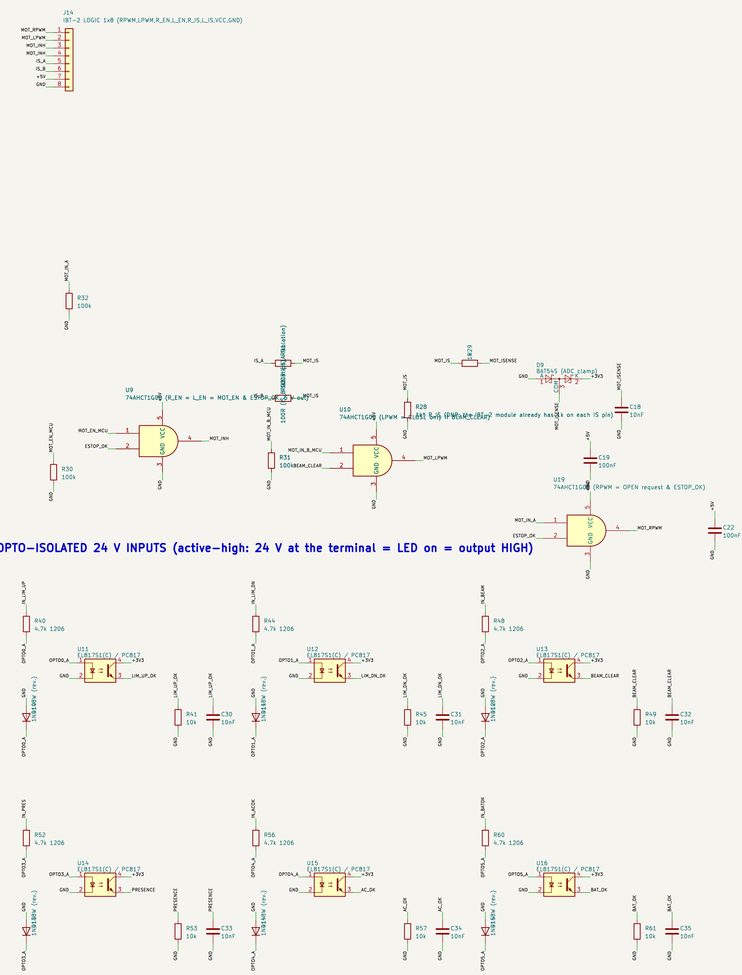
4. ESP32-S3-WROOM-1-N16R8 + RESET / BOOT / WATCHDOG / EDGE LINK



5. RS485 TO UHF RFID READER (THVD1450 + SM712)

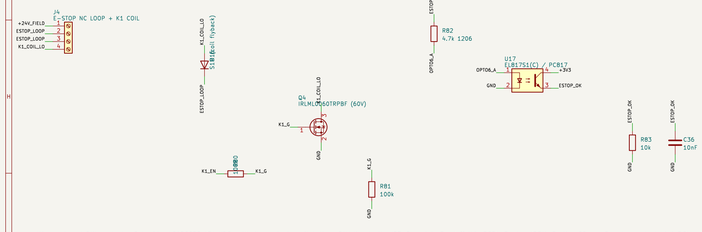


6. H-BRIDGE: IBT-2 (BTS7960) MODULE HEADER + hardware interlocks (74AHCT1G08 @ 5 V)



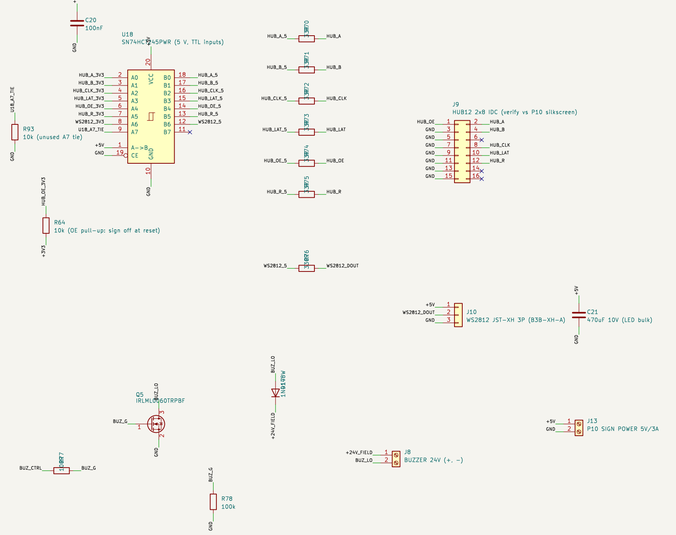
7. OPTO-ISOLATED 24 V INPUTS (active-high: 24 V at the terminal = LED on = output HIGH)

8. E-STOP LOOP + CONTACTOR COIL DRIVER (firmware-independent motor cut)

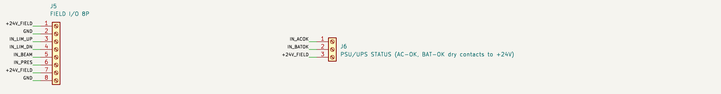


K1 = external DIN contactor Schneider LC1D098D (24 VDC coil): its contacts switch the +24V line to the IBT-2 module B+ (off this PCB). E-stop (BOM: ProMax 22 mm 2-pole, contacts 1 A DC) NC contact in series with the coil supply; pressing it de-energises K1 and removes motor power regardless of firmware. ESTOP_OK also forces the IBT-2 R_EN/L_EN and RPWM low via U9/U19. The contactor coil current (about 0.2 A at 24 V, assumed, to be confirmed on the datasheet) is carried by J4.1 / the E-stop contact, which is rated 1 A DC.

9. HUB12 (P10) + WS2812 LEVEL SHIFT (74HCT245 @5 V), BUZZER



10. FIELD CONNECTORS (5.08 mm fixed screw terminals / JST-XH)



J5: limit switches (BOM: ME-8108) wired on the NC contact between +24V_FIELD (J5.1) and IN_LIM_x (open = limit reached or wire break -> LOW -> safe): confirm the NC terminal on the real part.
IN_BEAM: the BOM through-beam receiver (Delta PS-M75 series, PNP) drives 24 V onto IN_BEAM when the beam is clear (light-ON) -> compatible with the active-high input: break / loss of supply = LOW = blocked (fail-safe). Confirm light-ON/NO output on the real part.
J6: dry contacts closed to +24V_FIELD (J6.3) = OK; no AC-OK / BAT-OK source device is in the BOM (LRS-150/200-24 has no status output): link J6.1 to J6.3 to disable mains-loss sensing in the baseline.

Notes: (a) Symbol MAX481E used for THVD1450DR (identical 8-pin RS485 pinout). (b) TLV62568DBV symbol used for TLV62569DBVR (same pinout).
(c) 74LS245 symbol used for SN74HCT245PWR and 74LVC1G08 symbol used for the 74AHCT1G08 gates U9/U10/U19 (same pinout).
(d) K1 is the external DIN contactor LC1D098D (24 VDC coil): only its coil terminals reach this PCB. The motor power path (K1 contacts -> IBT-2 module B+/B- -> motor) does NOT pass over this PCB.
(e) Values marked * are design targets -- verify against datasheets during layout. (f) GPIO3/45/46 (straps) and GPIO35-37 (octal PSRAM on N16R8) left unconnected; module is the BOM part N16R8, the N8 is footprint- and pin-compatible.

Design targets -- to be bench-verified
Generated by electronics/tools/gen_lane_controller_sch.py
Lane Controller PCB 2-layer ~100x80 mm
ParkFlow — Vishwakarma Awards Stage 2

Sheet: /	
File: lane_controller.kicad_sch	
Title: ParkFlow Lane Controller (ESP32-S3) — Rev A	
Size: A1	Date: 2026-10-24
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Rev: A
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